Finite singularities of the second-order Sobolev system are necessarily nonoscillatory

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13 August 2026

Abstract

Boulton and Lang conjectured that every finite singularity of the nonlinear four-dimensional system associated with second-order one-dimensional Sobolev embeddings is nonoscillatory, for all $p, q > 1$. We prove the conjecture, in fact for arbitrary initial states. The system has a conserved Hamiltonian which, at a zero of either derivative component, is a sum of two nonnegative potentials. Accumulation of such zeros would therefore bound both position components at all extrema; the differential equations then bound the entire state and permit continuation through the alleged singularity. Rolle's theorem excludes accumulating zeros of the position components. All signs are consequently fixed near a finite singularity, the equations force monotonicity, and boundedness of any one component propagates around the four-equation cycle. Thus every component tends in modulus to infinity and the two u-components have the signs opposite to the corresponding w-components, exactly as conjectured.

1 Exact source conjecture

For $r > 1$, write

$$\Phi_r(s) = \operatorname{sgn}(s)|s|^{r-1}, \qquad p' = \frac{p}{p-1}.$$

The source introduces

$$u_1 = u, \quad u_2 = u', \quad w_1 = -\Phi_p(u''), \quad w_2 = -(\Phi_p(u''))'$$

and the equivalent system

$$\begin{aligned} u_1' &= u_2, \quad u_2' = -\Phi_{p'}(w_1), \\ w_1' &= w_2, \quad w_2' = -\lambda\Phi_q(u_1), \end{aligned} \tag{1}$$

where $p, q > 1$ and $\lambda > 0$. The shooting problem in the paper starts at $(u_1, u_2, w_1, w_2) = (0, \alpha, 0, \beta)$, but the conjectural singularity statement is local and the argument below does not need this restriction.

After proving that an unbounded finite-time solution has either one-sign or highly oscillatory behavior, the paper states:

> "The solution does not become oscillatory near a singularity $t_\infty < \infty$ for a range of parameters (p, q). Namely
>
> $$\lim_{t \to t_\infty^-} |u_k(t)| = \lim_{t \to t_\infty^-} |w_k(t)| = \infty$$
>
> and (11) holds true. In fact we conjecture that the latter is the case for all $p > 1$ and $q > 1$, but we are not currently able to complete the proof of this claim."

Here equation (11) is

$$\lim_{t\to t_\infty^-} u_k(t) = -\lim_{t\to t_\infty^-} w_k(t), \qquad k=1,2, \tag{2}$$

as an equality of extended-real limits. Thus it asserts opposite blow-up signs, not equality of blow-up rates. The exact source display is reproduced below.

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2 The hidden first integral

The proof is driven by one identity not recorded in the source discussion.

Lemma 1 (Hamiltonian). *Every C^1 solution of (1) conserves*

$$H = u_2 w_2 + \frac{|w_1|^{p'}}{p'} + \frac{\lambda |u_1|^q}{q}. \tag{3}$$

Proof. For every $r>1$, the function $|s|^r/r$ is C^1 on $\mathbb{R}$ and has derivative $\Phi_r(s)$. Consequently,

$$\begin{aligned} \frac{dH}{dt} &= u_2' w_2 + u_2 w_2' + \Phi_{p'}(w_1) w_1' + \lambda \Phi_q(u_1) u_1' \\ &= -\Phi_{p'}(w_1) w_2 - \lambda u_2 \Phi_q(u_1) + \Phi_{p'}(w_1) w_2 + \lambda \Phi_q(u_1) u_2 = 0. \end{aligned}$$

This calculation remains valid when $p'<2$ or $q<2$; no second derivative of a power is used. $\square$

At any zero of u_2, identity (3) becomes

$$H = \frac{|w_1|^{p'}}{p'} + \frac{\lambda |u_1|^q}{q}. \tag{4}$$

The same equality holds at any zero of w_2. Thus either $H<0$ and neither derivative component ever vanishes, or $H \ge 0$ and both position components are uniformly bounded at every zero of either derivative component. This is the rigidity which prevents oscillatory blow-up.

3 No zeros can accumulate at a finite singularity

We first isolate the continuation fact used below. The right-hand side of (1) is continuous on $\mathbb{R}^4$. If the state is bounded on a finite interval $[t_0, T)$, then its derivative is bounded as well. The state is therefore uniformly Lipschitz, has a finite limit at T, and a local Cauchy–Peano solution starting at that limit concatenates with the original solution. Hence a nonextendible finite maximal endpoint requires an unbounded state. Uniqueness at the limiting state is not needed for this continuation argument.

Lemma 2 (zero-accumulation obstruction)**.** *Let a solution of* (1) *have a finite, nonextendible maximal right endpoint T. None of u_1, u_2, w_1, w_2 has zeros arbitrarily close to T.*

Proof. Suppose first that u_2 has zeros accumulating at T. At every such zero, (4) bounds $|u_1|$ and $|w_1|$ by constants depending only on H, p, q, and λ.

Choose one zero $s_0 < T$. Given $t \in [s_0, T)$, choose a later zero $s > t$. On the compact interval $[s_0, s]$, every maximum and minimum of u_1 occurs at an endpoint or at a point where $u_1' = u_2 = 0$. Its value is bounded in either case. It follows that u_1 is bounded on $[s_0, T)$, without any assumption that the zeros are isolated.

Now use the equations cyclically. Since u_1 is bounded, $w_2' = -\lambda\Phi_q(u_1)$ is bounded; on a finite interval this bounds w_2. Then $w_1' = w_2$ bounds w_1, and $u_2' = -\Phi_{p'}(w_1)$ bounds u_2. The whole state is bounded, so the solution extends through T, a contradiction. Therefore zeros of u_2 cannot accumulate at T.

The argument for w_2 is symmetric in structure. At every zero of w_2, (3) bounds u_1 and w_1. Extrema on intervals bracketed by such zeros show that w_1 is bounded on a tail. Then, successively, $u_2' = -\Phi_{p'}(w_1)$ bounds u_2, $u_1' = u_2$ bounds u_1, and $w_2' = -\lambda\Phi_q(u_1)$ bounds w_2. Again continuation contradicts maximality.

Finally, if zeros of u_1 accumulated at T, Rolle's theorem between successive zeros would produce zeros of u_2 accumulating at T. The same argument sends accumulating zeros of w_1 to accumulating zeros of w_2. Both possibilities have already been excluded. $\square$

The extremum argument in this proof is why neither simplicity nor discreteness of the zeros has to be assumed. It also covers a zero set containing an interval.

4 Full resolution of the conjecture

Theorem 1 (finite singularities are nonoscillatory)**.** *Fix $p, q > 1$ and $\lambda > 0$. Let (u_1, u_2, w_1, w_2) be any solution of* (1) *with a finite, nonextendible maximal right endpoint T. Then there are $t_0 < T$ and $\sigma \in \{-1, 1\}$ such that, for $t \in (t_0, T)$,*

$$\operatorname{sgn} u_1(t) = \operatorname{sgn} u_2(t) = \sigma, \qquad \operatorname{sgn} w_1(t) = \operatorname{sgn} w_2(t) = -\sigma.$$

Moreover,

$$\lim_{t\to T^-} |u_k(t)| = \lim_{t\to T^-} |w_k(t)| = \infty, \qquad k = 1, 2.$$

In particular (2) *holds, and the conjecture in arXiv:2204.04703 is true for all $p, q > 1$.*

Proof. By Lemma 2, after moving t_0 toward T if necessary, none of the four components vanishes. Each therefore has a fixed sign on (t_0, T). The four equations now show that every component is monotone there: u_1' and w_1' have fixed signs because u_2, w_2 do, while u_2' and w_2' have fixed signs because w_1, u_1 do.

We next show that no component can be bounded. Boundedness propagates around the system on a finite interval:

- if u_1 is bounded, then w_2' is bounded, hence w_2, then w_1, then u_2' and u_2 are bounded;
- if u_2 is bounded, then u_1 is bounded and the preceding chain applies;
- if w_1 is bounded, then u_2' is bounded, hence u_2, then u_1, and then w_2 are bounded;
- if w_2 is bounded, then w_1 is bounded and the preceding chain applies.

Thus boundedness of any one component would bound the whole state and permit continuation through T. Every component is therefore unbounded. Since it is monotone on a tail, its absolute value tends to infinity.

Let σ be the eventual sign of u_1. Since u_1 diverges to $\sigma\infty$ and $u_1' = u_2$ has a fixed sign, u_2 has sign σ. Next, $w_2' = -\lambda\Phi_q(u_1)$ has sign $-\sigma$. The monotone unbounded function w_2 must consequently tend to $-\sigma\infty$. Finally, $w_1' = w_2$ and unboundedness of w_1 force w_1 to have the same sign as w_2, namely $-\sigma$. This is the asserted sign pattern. $\square$

5 Checks, scope, and novelty audit

The proof has been audited against the following possible failure modes.

1. *Hamiltonian sign.* If $H < 0$, equation (4) says there are no zeros of either derivative component, so the argument only becomes easier. If $H \geq 0$, it supplies uniform bounds at all such zeros.

2. *Nonisolated zeros.* Compact-interval extrema, rather than a listing of consecutive simple zeros, give the tail bound in Lemma 2.

3. *Exponents below two.* The functions $|s|^r/r$ are C^1 for every $r > 1$, and the vector field is continuous. Neither the conservation law nor Cauchy–Peano continuation uses local Lipschitz continuity at zero.

4. *Initial data.* No step uses $u_1(0) = w_1(0) = 0$; the theorem is valid for arbitrary finite initial states.

5. *Meaning of equation (11).* The theorem proves opposite signs of the extended-real limits. It does not claim equal asymptotic magnitudes, which would generally be incompatible with the natural power-law scaling when $q \neq p'$.

The source paper, its published version, the run's cheap result indexes, and targeted searches for later work on this exact conjecture did not reveal a published resolution. Benedikt's earlier paper on the same initial-value equation gives explicit one-sign finite blow-up when $q > p$, but does not derive the Hamiltonian obstruction to oscillatory finite blow-up. This is a novelty check, not a guarantee of priority. Before dissemination, a specialist should independently verify the continuation convention and search MathSciNet/Zentralblatt citations for an equivalent first-integral argument.

References

[1] L. Boulton and J. Lang, *The eigenvalues and eigenfunctions of the non-linear equation associated to second order Sobolev embeddings*, Nonlinear Analysis 236 (2023), 113362; arXiv:2204.04703.

[2] J. Benedikt, *Uniqueness theorem for p-biharmonic equations*, Electron. J. Differential Equations 2002 (2002), no. 53, 1–17.